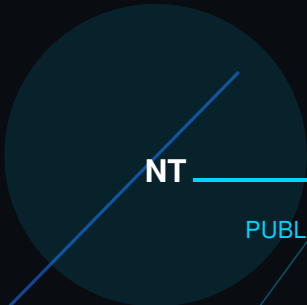




Node: The Building Block of NervTech Solutions

Initial Project Overview

A plain-language briefing for stakeholders, collaborators, and portfolio reviewers

PROJECT

Node

Document ID	NT-N-OVR-000	Revision	A
Status	Draft Brief	Date	2026-06-25
Audience	Non-technical stakeholders; Portfolio reviewers	Owner	NervTech Systems Engineering
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Purpose

In order to produce more complex platforms to test reinforcement learning and advanced controls research, a robust, multipurpose actuator must be developed that has high precision, bandwidth and torque capabilities. The Node fills this need, while being developed in house without the extra integration time and financial cost incurred by OEM integrated actuators.

Summary

This overview will define the scope, goals and key deliverables, including timelines, of the Node project. This is essential in ensuring project success is met in a desirable timeframe.

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1 PROJECT SNAPSHOT

Project Snapshot

What it is	A compact, easily integratable robotic actuator module that combines motion generation, sensing, electronics, and control in one reusable unit.
Why it matters	Robotic systems become easier to build when the joint actuator is robust, reusable and self-contained in terms of control and sensing.
Current stage	The project is in early architecture and prototype planning.
Next visible result	A 3D model of the first prototype in a simulated environment including the real life kinematics and dynamics of the system.

Background

As the labour automation industry becomes more expansive, older notions of developing highly specialised robotic systems for specific industries are becoming obsolete. As tasks with inherent uncertainty are being targeted for automation, distributed, easily configurable robotic systems will form the bedrock of the next generation of industry. NervTech systems aims to be at the forefront of this movement, and the Node project is a key step in that direction. The project will develop a reusable, highly versatile and easily integratable robotic actuator module that can be used in a variety of NervTech robotic systems. The eventual goal will be to develop a family of robotic components, including multi-dof arms and end effectors, that can be easily configured to form a variety of robotic systems for different applications.

2 PROBLEM STATEMENT

NervTech requires a compact, modular actuator that can provide highly precise controlled robotic motion for use in future distributed robotic platforms. The actuator must integrate sensing, control, communication, and mechanical mounting into a single reusable unit. Without such a platform, future robotic systems would require repeated custom integration of motors, sensors, drivers, and control hardware, increasing development time and reducing design consistency.

3 GOALS

Table 1: Overall Project Goals

NT Goal ID	Project Goal	Success Criteria	Value Target
NT-N-G1	The actuator should provide high precision torque and position control in high motion frequency applications.	The actuator should be able to move standard payloads up to 20kgs at frequencies comparable to human motion.	Precision and Versatility
NT-N-G2	The actuator should be reusable and easily integratable into a variety of frames for robotic systems.	The actuator should be able to be mounted and dismounted with conventional tools from any of the standard NervTech linkages and/or frames.	Reusability and Versatility
NT-N-G3	The actuator should be able to provide a variety of sensing capabilities for advanced control.	The actuator should be able to provide position, velocity, torque, and temperature sensing capabilities with minimal noise.	Control and Versatility
NT-N-G4	The actuator should be able to provide a variety of communication capabilities with minimal latency.	The actuator should be able to communicate with a variety of control systems using at least 2 standard industrial communication protocols, which can robustly deliver packets with sub-millisecond latency.	Control and Communication
NT-N-G5	The actuator should be able to be powered by standard LV external DC sources.	The actuator should be able to be powered via USB-C or standard molex connectors, from power supplies ranging from 24V - 48V.	Versatility and Power
NT-N-G6	The actuator should include standard industrial safety features.	The actuator should be able to provide safety features such as overcurrent protection, overtemperature protection, Safe Torque Off voltage lines and programmable max torque thresholds.	Safety and Reliability
NT-N-G7	The actuator should be durable and reliable for long term use.	The actuator should be able to operate for at least 10,000 hours of continuous operation without failure, and should be able to withstand standard industrial environmental conditions including dust and high moisture.	Reliability and Durability

4 DELIVERABLES

Table 2: Key Deliverables

Deliverable ID	Deliverable	Description	Deadline
NT-N-D1	Initial project overview	Plain-language document defining the product need, goals, requirements, success criteria, and roadmap.	4/07/2026
NT-N-D2	Preliminary research reports	Comparison of actuator housing architectures, speed reducer architectures, power control architectures, communication protocols, safety feature implementations and motion controller designs.	15/07/2026
NT-N-D3	Technical design documents	Mechanical, electrical, firmware and verification design documents, plus simulated environment using CAD, for the Node actuator.	26/07/2026
NT-N-D4	Prototype test report	Bench-test evidence including motion response, repeatability, current draw, fault behaviour, and lessons learned.	21/08/2026